Negativity bias in the cortical tracking of emotional representations in naturalistic stories

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Research has shown that negative meanings usually attract more attention than positive meanings during word and sentence evaluations, termed as negativity bias [1-3]. Such effect has been demonstrated in behavioral and electrophysiological (EEG/ERP) data. However, most evidence comes from controlled single-word or sentence paradigms to maximize emotion effects, leaving open how positive and negative information are tracked during continuous, naturalistic speech. Moreover, past ERP studies have largely adopted a lexical view for emotional valence of a word, without taking context into consideration. It is unclear whether emotional valence in natural speech is primarily lexically determined or contextually constructed, and how the span of contextual information modulates the emotional valence of a word in speech. Here we investigated how emotional valence of a word is tracked in the brain during naturalistic story comprehension, by directly comparing lexical (static) vs. contextual (dynamic) models of emotional valence in a continuous neural tracking framework.

We analyzed the Alice EEG dataset in which 33 participants listened passively to a 12.4-minute English audiobook recording of the first chapter of Alice's Adventures in Wonderland (2129 words) while their EEG was recorded [4]. Multivariate temporal response functions (mTRFs) were applied to estimate subject-specific linear kernels to predict the EEG data [5]. Models were estimated using acoustic (envelope, onset), lexical (word onset, log frequency, length, surprisal derived from the gpt-2 medium model), semantic (word-by-word semantic dissimilarity derived from the word2vec model), and emotional (positive valence, negative valence) predictors. Lexical valence was extracted from the Affective Norms for English Words (ANEW) [6], whereas contextual valence was estimated using the RoBERTa-base sentiment model [7], computed over prior 10-, 50-, 200-, and 1024-word tokens to probe the integration of different context spans. Model comparisons were conducted to assess (1) each predictor's unique explained variance (det > 0) as the detection-level analysis, and (2) incremental predictive gain from adding emotional predictors to the acoustic–lexical–semantic baseline.

Our detection-level analyses revealed widespread cortical tracking of negative contextual valence across all context spans (10 tokens: $p < .001$; 50 tokens: $p = .012$; 200 tokens: $p = .002$; 1024 tokens: $p < .001$). Positive contextual valence emerged only at an intermediate span (50 tokens; $p = .024$). No lexical valence effects were observed (Fig. 1A). However, stringent block-level model comparisons controlling for shared variance indicated that contextual valence at each span did not significantly improve model performance beyond the baseline model. Additionally, no significant differences were found when directly comparing contextual versus lexical valence, different context spans against one another, or the valence-specific unique contributions at each context span, over and above the baseline model. A follow-up exploratory analysis showed that adding negative contextual valence at the 1024-token span improved predictions over a baseline model including positive contextual valence ($p = .029$), although this effect did not survive FDR correction ($p = .116$; Fig. 1B), or appear at other spans.

Taken together, these findings suggest that emotional valence representations in naturalistic stories are predominantly context-dependent rather than lexically driven. While our detection analyses indicate general neural sensitivity to negative contextual information, incremental model comparisons reveal only statistically weak predictive gains, limited to the longest discourse span (1024 tokens). Our results therefore provide initial evidence for contextual affective tracking in continuous speech, without strong support for a robust negativity bias under stringent variance-partitioning criteria.

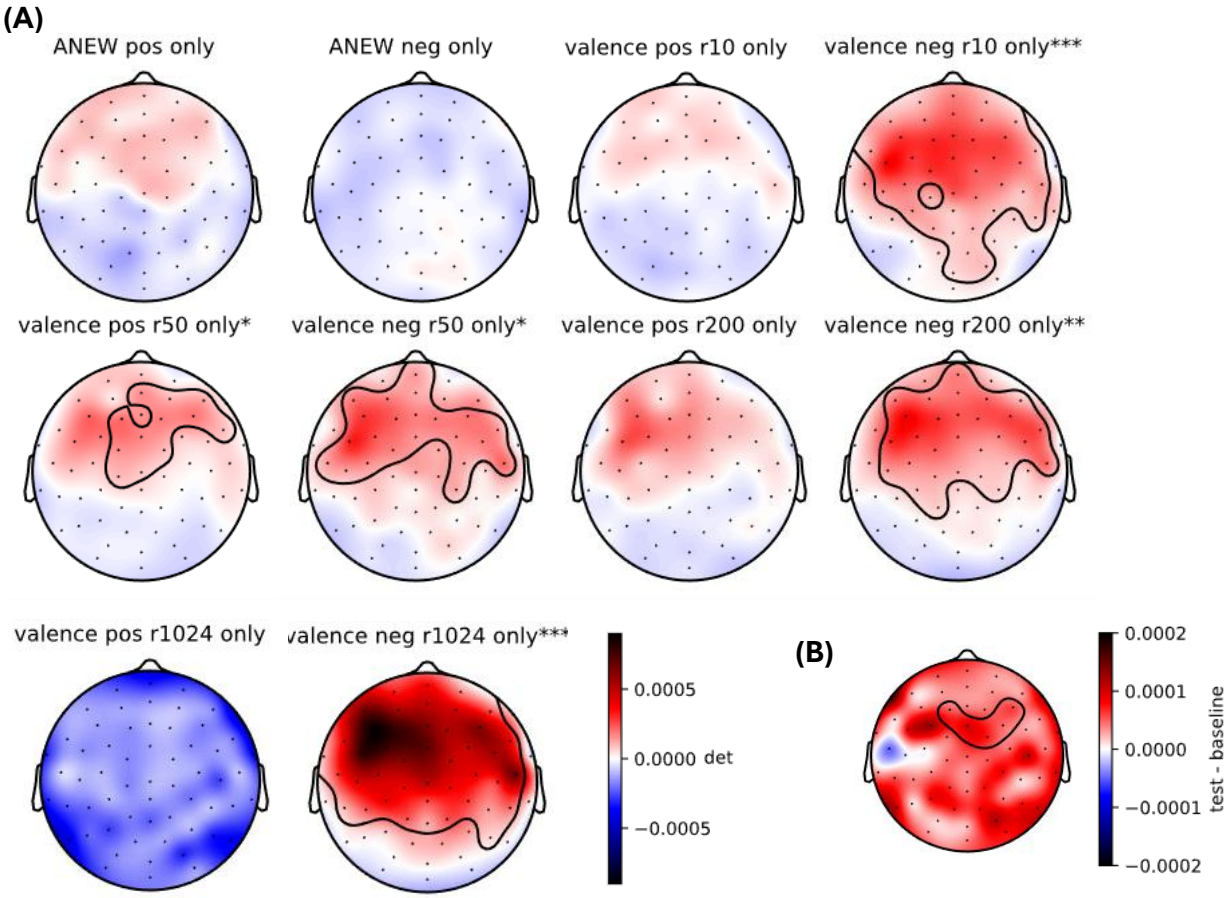

Figure 1. (A) Model comparisons of predictive power (det = explained variance) against zero for each lexical (ANEW positive valence, ANEW negative valence) and contextual valence (r10/r50/r200/r1024 = 10/50/200/1024-token span) across the scalp. Each plot shows a head-map of the change in predictive power. Significant results were marked on the title (*p < .05; **p < .01; ***p < .001). **(B)** Head-map of the change in predictive power between the test model (acoustic envelope + acoustic onset + word onset + word log frequency + word length + word surprisal + semantic dissimilarity + RoBERTa_negative_r1024) vs. baseline (acoustic envelope + acoustic onset + word onset + word log frequency + word length + word surprisal + semantic dissimilarity + RoBERTa_positive_r1024) model. The legend units stand for the improvement of explained variance (Δdet); Black outlines mark significant clusters, p ≤ 0.05, family-wise error corrected for the whole head map

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